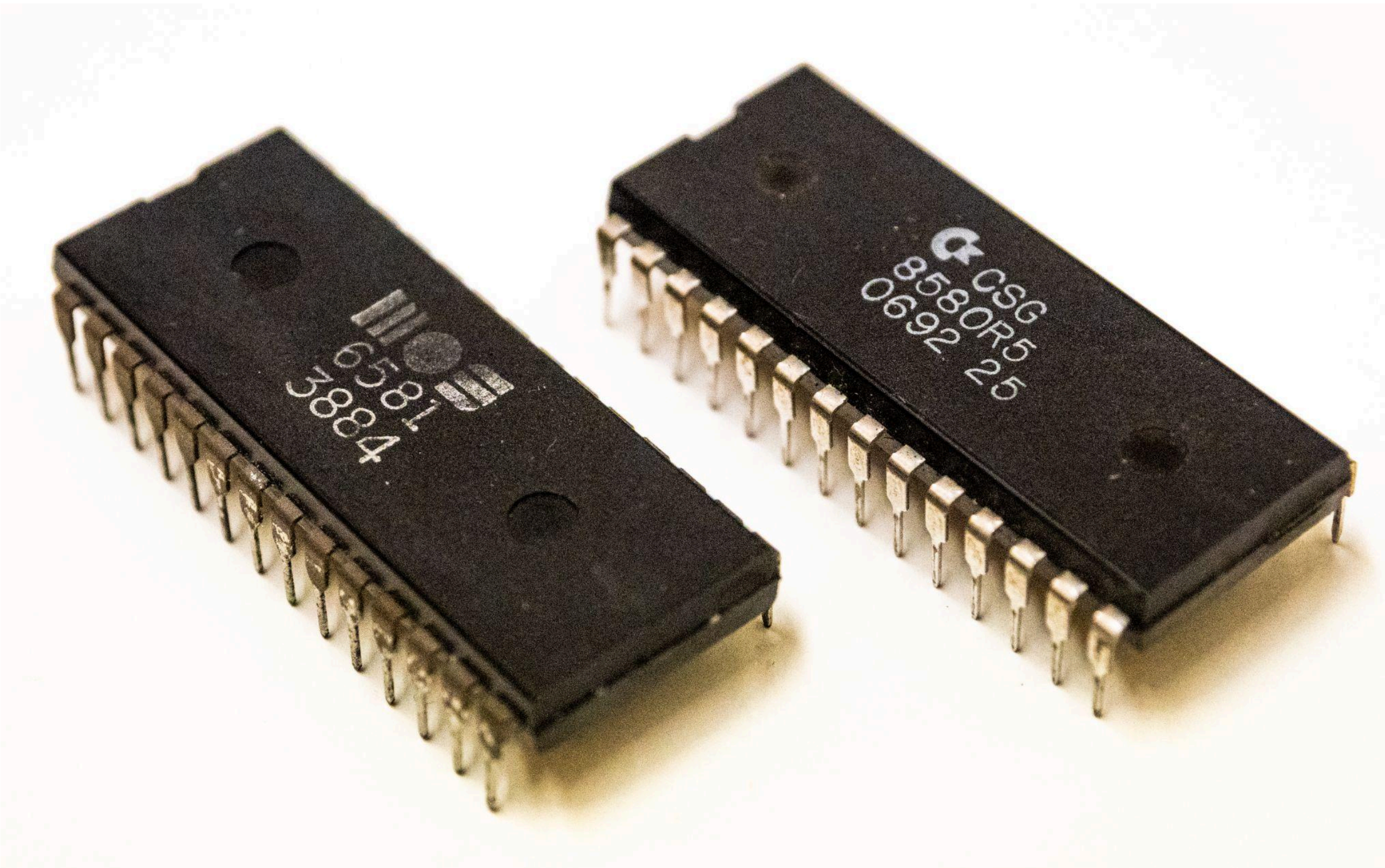


"Double your Listening Pleasure"

One of the coolest features of the EVO is its native support for Dual SID chips and stereo sound output. The original inspirations came from the **MixSID by Henning Liebenau**, although EVO takes a completely unique and novel approach by implementing all of its dual-SID addressing logic directly into a customized PLA we're calling the **QAPLA'**.

You may be familiar with some of the popular dual-SID solutions out there, which typically send the same (*mono*) audio signal to both SIDs at the same time. This can produce a nice "pseudo-Stereo effect", when different types of SID chips are used (e.g., one 6581 and one 8580). This is due to the fact that your ears are hearing the different sound characteristics from the different chip models. EVO can reproduce this effect too of course, but it also goes *much* further…

EVO provides “True Stereo Sound”, by exposing the second SID to its own *unique* memory addresses. This enables different notes and effects to be played from each SID, providing software with access to 6x oscillator voices and 2x filters, for *double the fun*.

To ensure the best backwards compatibility, the primary SID on your EVO will *always* listen on address $D400, and by moving a simple jumper, you can select which of your two SIDs will act as the primary at any given time. By default, your second SID will receive notes and instructions from *both* $D420 and $D500, which enables great support for most of the popular dual-SID tracks in the **HVSC**, as well as many of the great dual-SID capable games, like the new Mario Brothers. You can also make your own stereo SID music, with composer software like the **CYNTHCART**. Moving another jumper will allow the second SID to listen on $DE00 and $DF00 as well, which provides even further compatibility for tools like the **MSSIAH** music composition cartridge, and other music players.

Best of all, EVO lets you use *any* SID chip variant(s) that you prefer, from the earliest generation 6581, with its gnarly oscillators and filter glitches (operating at 12v), to the clean and pristine 8580 (operating at 9v). And as you would expect, EVO is also fully compatible with the most popular SID clones, like the **ARMSID**, the **FPGA SID**, the **SwinSID** and the **SwinSID Ultimate**, the **BackSID**, etc. Best of all, you can mix and match any combination of SIDs that you like, to get the best version of that retro sound you’ve been hearing in your head.

Some suggestions for enjoying SID Music:

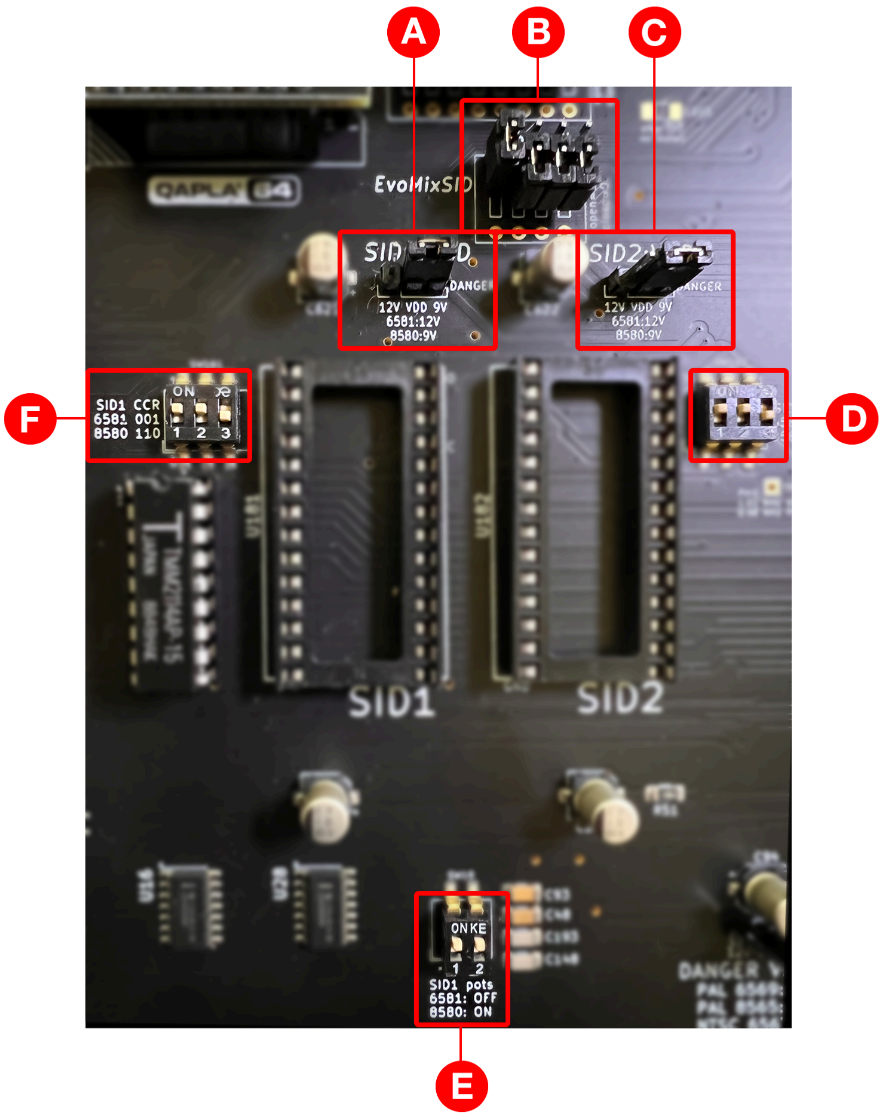
- Download the latest **High Voltage SID Collection (HVSC)**, which contains over ~57,500 SID tunes, spanning the entire history of the C64, including all your favorite games and demos
- Check this handy reference if you **just want to hear the dual-SID tracks**, organized by the year they were added into the HVSC
- You can enjoy your favorite SID tracks pretty easily if you happen to have an **Ultimate II+ Cartridge**, or the **SIDEKick**, which have their own built-in SID player apps
- You can also download **SIDPlay64**, which comes in a variety of different versions for accessing different types of storage devices. There’s a version that works well with standard disks, one for the SD2IEC device, one for use with a Ram Expansion Unit (REU), etc., etc.

The following pages provide the necessary configuration steps to help you get your EVO64 system setup to support the SIDs (or clones) of your choice, with multiple options for audio output signal routing.

Please enjoy!

Jumper Settings for Dual-SID Control

The **quick reference table** below provides information to help with jumper configurations and DIP switch settings that are needed to ensure that your SID chips are working properly and providing their best quality audio output, as well as full support for analog devices like paddles and a mouse



A & C SID Voltage Selection*

12V	All 6581 model SIDs
9V	8580, 6582 and all clones

B Multi-SID Mode Selection

PIN	OPEN	CLOSED
MS	Dual Mono	Stereo
PS	Primary SID1	Primary SID2
IE1	Disable $DE00	Enable $DE00
IE2	Disable $DF00	Enable $DF00

D & F SID Filter Capacitor Type

SID Type	1	2	3
6581	off	off	ON
8580 / 6582	ON	ON	off

E SID Paddle Capacitor Type

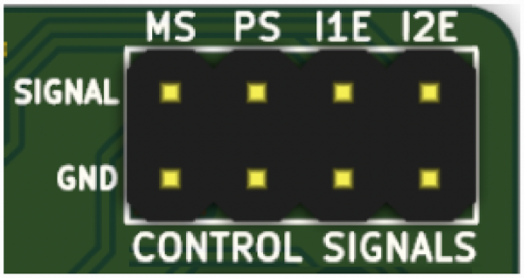
SID Type	1	2
6581	off	off
8580 / 6582	ON	ON

IMPORTANT

- **Never make changes to voltages while your system is powered on!**
- **Sending 12v into a 8580 could fry it (quickly). Be sure to double and even triple check your voltage settings *before* powering up your system**
- **Your SID chips are likely to sound terrible if you select the wrong filter capacitor setting (D & F above), so ensure that you have the correct type selected whenever you chance SID chips**

EvoMixSID Array Bank

The next section provides a bit more insight into the various multi-SID operations that EVO supports.

	**NOTE:** All runtime control signals are pulled up to "logic HIGH" internally when no jumper is connected. In other words, the top row contains the actual signals, while all pins on the bottom row are all connected to ground (GND) These pins can either be left unconnected (HIGH) or connected to GND (LOW). These signals can be freely changed at runtime, without worrying about crashing the system, so *feel free to experiment with different settings while the machine is turned on and playing music!*

Mode Select (MS)

The mode select line switches between **parallel** or **stereo** addressing modes:

Parallel Addressing Mode (Jumper OPEN = HIGH)
This mode should be used when playing back normal SID tunes which expect just one SID at $D400. Both SIDs are addressed in parallel, so both are available on $D400 in this mode, and they'll both playback the same tune.

In this mode, the secondary SID is forced into write-only, while the primary SID remains both read and write. So, only the primary SID will respond to read requests to the readable SID registers (POTX, POTY, OSC3 and ENV3). This way paddles and mice as well as sound effects that are based on reading OSC3 or ENV3 will work as usual.

If you install SIDs that differ in sound (either a 8580 and a 6581, or two different 6581 versions with audible differences) and listen to each SID on a different stereo channel, you will notice the aforementioned Pseudo-Stereo effect, even while playing a simple mono tune. This results from the SID differences. Whether or not this sounds good to *you* is highly subjective, and will mostly depend on the tune itself.

Stereo addressing Mode (Jumper CLOSED = LOW)
This mode can be used when playing back tunes that were explicitly created for two SID chips.

When in stereo addressing mode, the primary SID will be addressed at $D400 while the secondary SID will be addressed at *both* $D420 and $D500 simultaneously. Most stereo tunes that were tested seem to expect the second SID at one-of-these-two addresses (e.g., a majority of the *2SIDs in the High Voltage SID Collection).

If the IO Enable signals are used, the secondary SID will *additionally* be addressed at $DE00 and/or $DF00. While fewer tunes expect the second SID at these addresses, some hardware extensions may want to see the second SID there (e.g., the MSSIAH Cartridge), however these addresses may conflict with some software and cartridges (e.g., the RetroReplay), so *it's best not to enable these addresses unless you know that you need them.*

Priority Select (PS)

The priority select signal controls which SID takes the role of primary and which takes the role of the secondary.

The primary SID will *always* be read-write at $D400. However, the secondary SID will either be write-only at $D400 when Mode Select is set to parallel addressing mode, or it will be full read-write when Mode Select is set to stereo addressing mode (see Mode Select for more details).

If this signal is HIGH (e.g, jumper OPEN), the SID in the *right* socket (labeled SID1) will be the primary SID. If the signal is LOW (e.g., jumper CLOSED), the SID in the left socket (labeled SID2) will be the primary SID.

Effects on software SID detection

A common method for detecting the model of the installed SID from within a C64 program involves reading the OSC3 and ENV3 registers.

Since only the primary SID will respond to read requests, the priority selection also controls which SID model is detected by software in a mixed setup.

Effects on paddle accuracy

The POTX and POTY lines will always be routed to the primary SID, which will then measure and report the current paddle position to the software reading the POTX and POTY registers.

The actual values measured by the SID depend on the value of the capacitors externally connected to the POTX and POTY lines .

Since different values were used on different mainboards (depending on the SID model usually used on that main board), the measurements will be slightly off if a SID model is used as the primary SID that is not native to the board.

For example, when using a 8580 as the primary SID on a 250407 mainboard, the values read from the paddle can be off by about ~5%.

However, this hardly matters in practice since the complete range of values can still be reached by turning the paddle knob. This is only a problem if the middle position of the paddle absolutely has to yield a value of 127. In this case, use at least one SID model that is native to your mainboard type, and set it to be the primary SID when using paddles.

IO Enable (I1E, I2E)

The I1E and I2E signals control whether or not the secondary SID also appears at the corresponding stereo addresses $DE00 or $DF00 when in stereo addressing mode (e.g., Mode Select [MS] must be set to Stereo Address Mode [Jumper Closed]).

HIGH (Jumper Open): 2nd SID does ***not*** appear at corresponding IO-Area address
LOW (Jumper Closed): 2nd SID ***does*** appear at corresponding IO-Area address

This feature allows the user to permanently connect the I/O signal lines to the board, but still control at runtime whether or not they are used for stereo mode.

SID Audio Amplifier Control

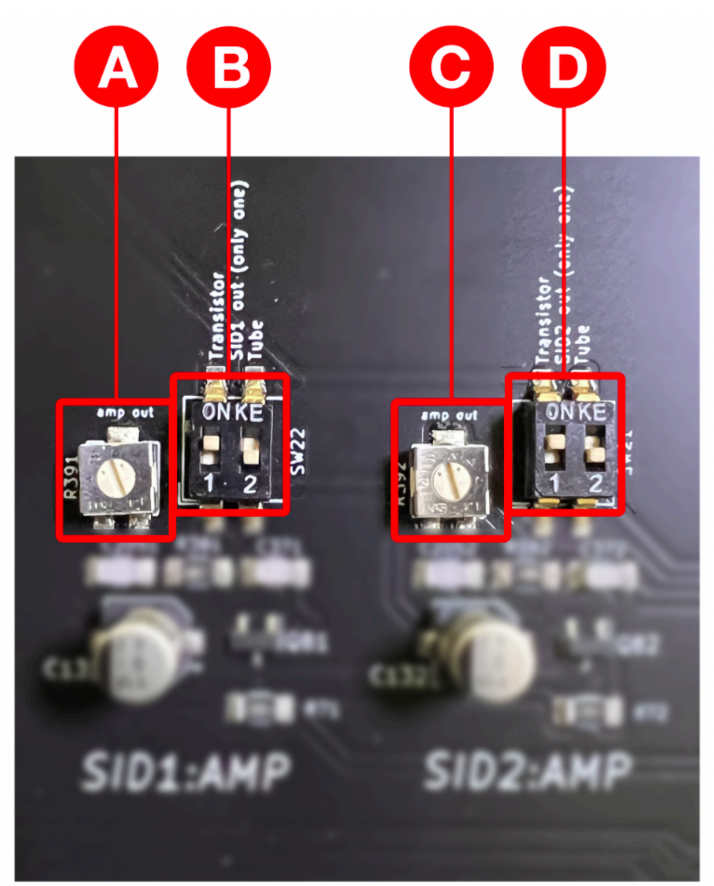
You have complete control over which Audio output signal(s) will flow through through your system's 8-Pin Audio / Video jack on the rear of your system. You can choose to leverage the built-in "transistor-based amplifiers" or if you happen to have one of the optional audio preamp modules (**[the NuTube64 or Triode64](#)**), you can direct their fully processed and colorized audio output through your A/V port.

Even if you have a preamp module, we think that most people will still choose to use the built-in transistor amps for their standard audio output that will flow into their monitor and/or speakers, since this will produce the type of line level audio output signal that those devices are expecting.

Using one of the 3.5mm (1/8th inch) jacks on the back of the preamp module is going to be the best way to capture the processed and warmed audio output, which you will most likely want to send into an amplifier, a sound card for a DAW or higher quality set of powered monitor speakers.

This same section of the EVO board also includes a set of potentiometers that allow you to make small adjustments to the gain levels for each of your SIDs independently. These ensure that you'll always have properly matched audio output levels from your left and right channels (stereo or dual-mono), even when you're mixing and matching with different types of SIDs models, which vary in their output levels.

The table below provides a **quick reference** to help configure the SID audio routing and output levels.



A & C SID Amplifier Gain Level

Use a small screwdriver to make fine adjustments to these potentiometers, which allow you to rebalance the gain levels for each of your SID chips independently. Audio is sent to the 8-Pin Audio / Video port on the rear of the system

TIP: It's best to make these adjustments while you are playing back SID music and listening closely to the audio output

B & D SID Audio Amplifier Selection

These DIP switches control which audio output (e.g., **internal amp vs. optional preamp add-on**) will be sent to the 8-Pin DIN Audio / Video port on the rear of the system*

SID AMPLIFIER TYPE	1	2
Internal (Transistor) Amp	ON	off
Optional Tube Preamp	off	ON

***Be sure to select only one audio routing option.**

Selecting the A/V Audio Output Pin for your Second SID

The original C64 systems didn't have multiple SIDs, so by default the built-in 8-Pin DIN Audio / Video (AV) jack on the back of a standard C64 system wouldn't have support for the second SID's audio output.

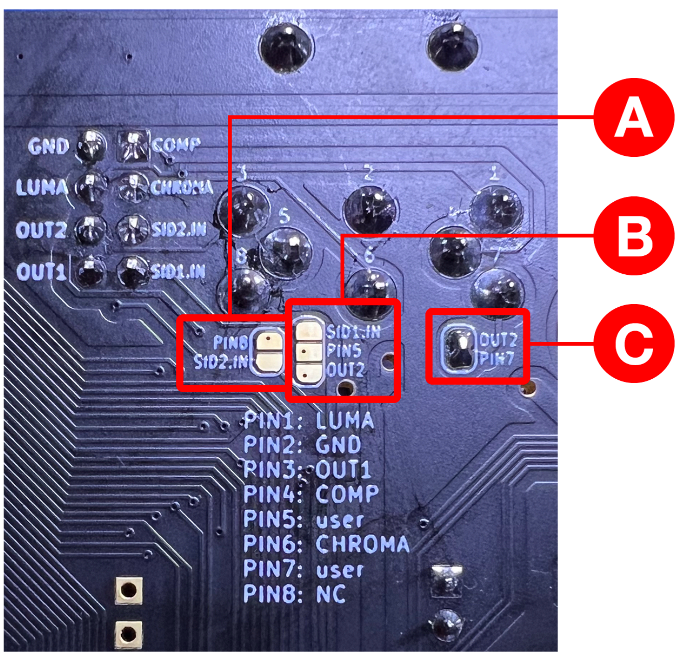
Fortunately EVO has you covered there, allowing you to select which of the unused AV jack pins you would prefer to use to capture audio from the second SID.

The most common standard nowadays is to use Pin-7, but some people have a preference for Pin-5, depending on the types of AV cables they may already have lying around.

NOTE: If you received a pre-assembled EVO, it will likely have already been configured to use Pin-7, so you can skip this section and avoid any soldering

The table below provides a **quick reference** to help configure a series of solder-pads on the **UNDERSIDE** of the EVO, which are located right below the Audio / Video port (AV). These pads control the audio input and output signal routing, in to and out from the AV port.

***IMPORTANT* This configuration step is *required* in order to be able to hear audio output from SID #2, irrespective of whether you plan to use Dual-Mono or Stereo mode**



A **Audio "Input" to SID #2 (optional)**

Shorting across these pads will enable you to send an external audio signal "IN" to your system over **AV pin 8**, which will be directed to the SID #2. **This feature is *not* recommended** as it could introduce noise into the system

B **SID #2 Audio "Output" to AV Pin 5***

OPTION #1: Shorting the **middle** and **bottom** pads will send **SID #2 audio output** to **AV pin 5**. You must have a custom AV cable that supports this (see the next section below)

OPTION #2: Shorting the **top** and **middle** pads will configure AV Pin 5 to accept an external audio signal "IN" to your system on AV Pin 5. **This feature not recommended** as it could introduce noise into the system

***Do *NOT* short all three of these pads, or try to combine options 1 & 2. This will cause audio distortion!**

C **SID #2 Audio "Output" to AV Pin 7**

Shorting these pads will send **SID #2 audio output** to **AV pin 7**, *which is the most common choice for most of the stereo AV cables that have been made available for the Commodore 64 over the years*

You need an A/V breakout module or a custom cable that supports dual-SID output (see next section)

TIP: Pin 7 is the most widely recommended option for capturing stereo audio output from the C64

Audio / Video Cables (AV)

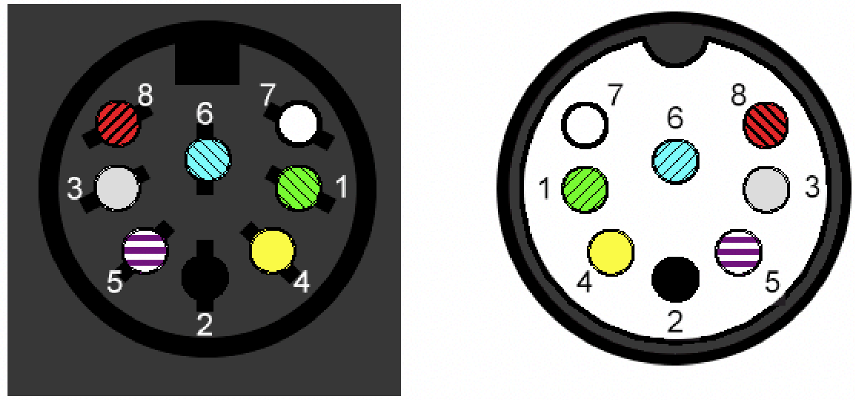
Most of the standard AV cables made for the C64 are not going to provide support for a second SID's audio output. This leaves you with a few choices to consider.

OPTION 1: You can buy a modern AV cable from a supplier like **8 Bit Classics**, which supports second-SID audio through Pin-7 of the AV jack (note: we are not affiliated with or compensated by them)

OPTION 2: You can pick up a nice breakout module from the **Retrospective shop**, which can plug directly into your system's AV Port and provide separate output signals for the left and right audio channels, as well as separate signals for composite and S-Video. This approach enables you to use standard off the shelf RCA and S-Video cables, which are widely available from many retailers.

OPTION 3: You can have your own AV cable(s) made, by following the specification below, but you'll need to be mindful of the details. The C64's 8-Pin DIN uses a "horseshoe shaped" configuration at a 262 degree angle. There are other 8-Pin connector standards with pins on different angles, *which won't fit your machine properly!*

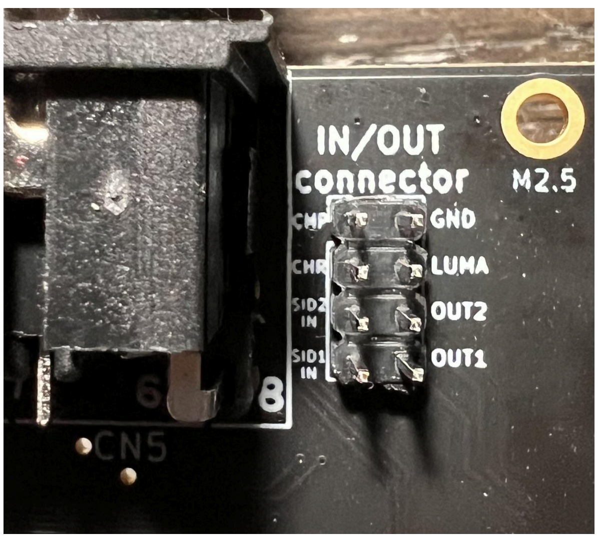
DIN-8 262

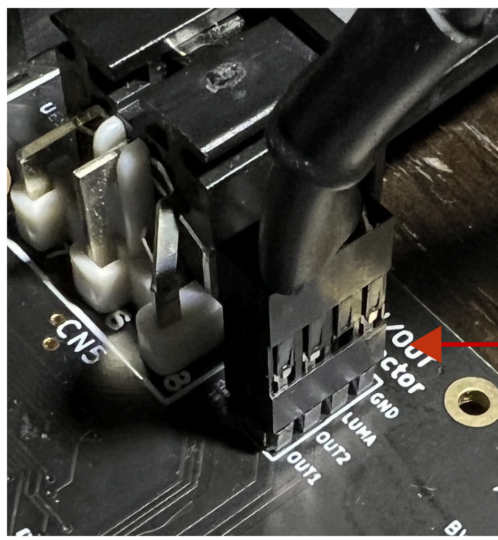


Female (Computer) Male (Cable)

1. S-Video Luma
2. Ground
3. Audio Right Output (Red RCA jack center)
4. Composite (Yellow RCA jack center)
5. Audio Input to SID #1
6. S-Video Chroma
7. Audio Left Output (White RCA jack center)
8. +5V

OPTION 4: You could choose to make a dedicated dual-channel audio-output only cable, which you can connect to your system's INPUT / OUTPUT connector, which is located to the right of you AV DIN





Stereo Audio Cable

For Audio Preamp Users

If you're planning to use any of the audio preamp modules being made available for EVO (e.g., the NuTube64 or the Triode64), please refer to **the appropriate documentation** for installation and setup, which will ensure that you're always having the best audio experience with them.

NuTube64 (Gen 3)	**Triode64** (Gen 1)